Topic 13: Chemical Equilibria

Knowledge of the concepts introduced in Unit 2, Topic 9B Chemical Equilibria will be assumed and extended in this topic.

Students will be assessed on their ability to:

13.1	be able to deduce an expression for K_c , for homogeneous and heterogeneous systems, in terms of equilibrium concentrations
13.2	be able to deduce an expression for K_p for homogeneous and heterogeneous systems, in terms of equilibrium partial pressures in atm
13.3	be able to calculate a value, with units where appropriate, for the equilibrium constants (K_c and K_p) for homogeneous and heterogeneous reactions, from experimental data
13.4	understand how, if at all, a change in temperature, pressure or the presence of a catalyst affects the equilibrium composition in a homogeneous or heterogeneous system
13.5	understand that the value of the equilibrium constant is not affected by changes in concentration or pressure or by the addition of a catalyst
13.6	know the effect of changing the temperature on the equilibrium constant (K_c and K_p) for both exothermic and endothermic reactions
13.7	understand that the effect of temperature on the position of equilibrium is explained using a change in the value of the equilibrium constant
13.8	understand the effect of a change in temperature on: i the value of ΔS_{total} ii the magnitude of the equilibrium constant, since $\Delta S_{total} = R \ln K$
13.9	be able to apply knowledge of the value of equilibrium constants to predict the extent to which a reaction takes place
	Further suggested practicals: i the reaction of ethanol and ethanoic acid (this can be used as an example of the use of ICT to present and analyse data) ii the equilibrium $Fe^{2+}(aq) + Ag^{+}(aq) \rightleftharpoons Fe^{3+}(aq) + Ag(s)$ iii the distribution of ammonia or iodine between two immiscible solvents iv the thermal decomposition of ammonium chloride v the effect of temperature and pressure changes in the system $2NO_2 \rightleftharpoons N_2O_4$